

**Writing Large Numbers in Scientific Notation**Question Bank 3.08 **ANSWER KEY**

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Show your work. Circle your final answer.

**For questions 1 through 4, write each number in scientific notation.**

1. 9,000

$$9 \times 10^3$$

2. 4,500,000

$$4.5 \times 10^6$$

3. 310,000,000

$$3.1 \times 10^8$$

4. 7,020,000,000

$$7.02 \times 10^9$$

*The zero between the 7 and the 2 stays in the coefficient.*

**For questions 5 and 6, write each number in standard form.**

5.  $6.3 \times 10^4$

$$63,000$$

6.  $1.05 \times 10^8$

$$105,000,000$$

**Real quantities.**

7. Write each quantity in scientific notation.

- a. There are about 1,670,000,000,000,000,000,000 molecules in a drop of water.

$$1.67 \times 10^{21} \text{ molecules}$$

- b. The average distance from Earth to the Sun is about 93,000,000 miles.

$$9.3 \times 10^7 \text{ miles}$$

- c. A computer displays the number of stars in a galaxy as **4E11**. Write it in standard form.

$$400,000,000,000$$

*E11 means  $\times 10^{11}$ .*

8. Complete each comparison.

- a. How many times larger is 8.1 than 5.4?

**1.5**

- b. How many times larger is  $10^8$  than  $10^7$  ?

**10**

- c. How many times larger is  $8.1 \times 10^8$  than  $5.4 \times 10^7$  ?

**$1.5 \times 10 = 15$  times larger**

9. Answer each question.

- a. A blue whale weighs about  $4 \times 10^5$  pounds. An elephant weighs about  $1 \times 10^4$  pounds. How many times heavier is the blue whale?

**40 times**

$4 \div 1 = 4$  and  $10^5 \div 10^4 = 10$ .

- b. Florida has an area of about  $6.5 \times 10^4$  square miles. Lake Okeechobee has an area of about  $7.3 \times 10^2$  square miles. About how many times larger is Florida? Round to the nearest whole number.

**about 89 times**

$6.5 \div 7.3 \approx 0.89$  and  $10^4 \div 10^2 = 100$ ;  $0.89 \times 100 \approx 89$ . Directly:  $65,000 \div 730 \approx 89.04$ .

10. Which of the following is greater? Explain how you know.

- a.  $9.5 \times 10^5$  or  $1.2 \times 10^6$

**$1.2 \times 10^6$  —  $10^6$  is greater than  $10^5$ , so the powers decide;  $1,200,000 > 950,000$ .**

- b.  $3.8 \times 10^7$  or  $3.09 \times 10^7$

**$3.8 \times 10^7$  — same power, so compare coefficients:  $3.8 > 3.09$ .**

11. Which is 52,000,000 written in scientific notation?

● **A.  $5.2 \times 10^7$**

○ B.  $5.2 \times 10^8$

○ C.  $5.2 \times 10^6$

○ D.  $52 \times 10^6$

**A**

12. Select ALL of the expressions that are equal in value to 3,400,000.

- ☐ A.  $3.4 \times 10^5$
- ☒ B.  $0.34 \times 10^7$
- ☒ C.  $34 \times 10^5$
- ☒ D.  $3.4 \times 10^6$
- ☒ E.  $340 \times 10^4$
- ☐ F.  $3.4 \times 10^7$

**B, C, D and E**

*Only D is in scientific notation; B, C and E are equal in value but their coefficients are not in  $[1, 10)$ .*

13. In question 12, only one of the four equal expressions is written in scientific notation. Which one is it, and why are the other three not?

**$3.4 \times 10^6$  . In scientific notation the coefficient must be at least 1 and less than 10; 34, 340 and 0.34 are not.**